

Bohr-lift zero incomparability

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Source and Target

The source paper is Jeet Sampat's arXiv:2409.10224 paper [1], *Shift-cyclicity in analytic function spaces*. The target is Problem 2.23 on PDF page 20. The paper defines the Bohr transform $\mathfrak{B} : H_0^2(\mathbb{D}) \rightarrow H^2(\mathbb{D}_2^\infty)$ by sending the coefficient of z^n to the monomial indexed by the prime factorization of n . The problem asks how $f(\mathbb{D})$ and $\mathfrak{B}f(\mathbb{D}_2^\infty)$ are related and, in particular, asks for a classification of those $f \in H_0^2(\mathbb{D})$ for which $\mathfrak{B}f$ is non-vanishing on $\mathbb{D}_2^\infty$.

Let $H_0^2(\mathbb{D})$ be as in Example 1.5 (2) and note that $L^2(0,1)$ and $H_0^2(\mathbb{D})$ are unitarily equivalent via

$$\sqrt{2}\sin(\pi nx) \leftrightarrow z^n.$$

Under the same correspondence, it is also clear that ϑ -cyclicity is equivalent to τ -cyclicity (again, from Example 1.5 (2)). Taking things one step further we can show that τ -cyclicity is equivalent to shift-cyclicity in $H^2(\mathbb{D}_2^\infty)$. Indeed, we can use an idea of Bohr [22] for this. Let us order all the prime numbers p_m in increasing order along $m \in \mathbb{N}$. Every $n \in \mathbb{N}$ has a unique prime factorization

$$n = p_1^{\kappa_1(n)} p_2^{\kappa_2(n)} \dots,$$

which gives us a bijective map $\kappa : \mathbb{N} \rightarrow \mathbb{Z}_+^\infty$ via

$$\kappa : n \mapsto (\kappa_1(n), \kappa_2(n), \dots).$$

We use this bijective map to obtain a correspondence, called the *Bohr transform*,

$$\mathfrak{B} : \sum_{n \in \mathbb{N}} c_n z^n \in H_0^2(\mathbb{D}) \mapsto \sum_{\alpha \in \mathbb{Z}_+^\infty} c_{\kappa^{-1}(\alpha)} z^\alpha \in H^2(\mathbb{D}_2^\infty).$$

It is a standard exercise to check that $\mathfrak{B} : H_0^2(\mathbb{D}) \rightarrow H^2(\mathbb{D}_2^\infty)$ is a unitary and τ -cyclicity is equivalent to shift-cyclicity in $H^2(\mathbb{D}_2^\infty)$ (see [83, Section 6.6.3]).

While things are clear from an operator theoretic point of view, notice that point-wise relationships are lost under these unitary equivalences. For instance, Nikolski [81] used a theorem of Green and Tao [43] about the existence of arbitrarily long arithmetic progressions among the prime numbers to show that

$$\mathfrak{B}(H^\infty(\mathbb{D})) \not\subset H^\infty(\mathbb{D}_2^\infty)$$

(also see [83, Exercise 6.7.4 (c)]). Moreover, from P2 it is clear that if F is shift-cyclic in $H^2(\mathbb{D}_2^\infty)$ then it is non-vanishing on $\mathbb{D}_2^\infty$, but it is not clear what this property corresponds to under $\mathfrak{B}^{-1}$. We therefore end this subsection with the following open problem.

Problem 2.23. Are $f(\mathbb{D})$ and $\mathfrak{B}f(\mathbb{D}_2^\infty)$ related to each other for any given $f \in H_0^2(\mathbb{D})$? In particular, can we classify all $f \in H_0^2(\mathbb{D})$ such that $\mathfrak{B}f$ is non-vanishing on $\mathbb{D}_2^\infty$.

Idea of the Proof

The first three positive indices already split under the Bohr transform: 1 has the empty prime factorization, while 2 and 3 become the first two coordinate variables. Thus the ordinary one-variable cubic

$$f(z) = az + bz^2 + cz^3$$

is turned into the affine two-variable function

$$\mathfrak{B}f(w) = a + bw_1 + cw_2.$$

Nonvanishing on the Hilbert multidisk is therefore controlled by whether the disk centered at a with radius $|b| + |c|$ reaches the origin. This gives a complete degree-three criterion and a simple example whose ordinary disk zeros do not predict Bohr-lift zeros. A second finite example, using the factorization $(1 - w_1)(1 - w_2)$, shows the reverse failure: Bohr-lift nonvanishing does not imply ordinary disk nonvanishing away from the forced zero at the origin.

Theorem 1. *Let*

$$f(z) = az + bz^2 + cz^3 \in H_0^2(\mathbb{D}).$$

Then $\mathfrak{B}f$ is non-vanishing on $\mathbb{D}_2^\infty$ if and only if

$$a \neq 0 \quad \text{and} \quad |a| \geq |b| + |c|.$$

Proof. Because 1 has the empty prime factorization, while $2 = p_1$ and $3 = p_2$, the Bohr transform is

$$\mathfrak{B}f(w) = a + bw_1 + cw_2, \quad w = (w_1, w_2, \dots) \in \mathbb{D}_2^\infty.$$

If $a = 0$, then $\mathfrak{B}f(0) = 0$, so nonvanishing is impossible.

Assume $a \neq 0$ and $|a| \geq |b| + |c|$. For every $w \in \mathbb{D}_2^\infty$,

$$|bw_1 + cw_2| < |b| + |c| \leq |a|,$$

where the first inequality is strict unless $b = c = 0$, in which case the claim is immediate. Hence $a + bw_1 + cw_2 \neq 0$.

Conversely, suppose $|a| < |b| + |c|$. If $a = 0$, the zero at $w = 0$ has already been noted. Otherwise choose $0 \leq r_1, r_2 < 1$ such that

$$|a| = |b|r_1 + |c|r_2,$$

with the evident convention that a term with zero coefficient is omitted. Choose phases for w_1, w_2 so that bw_1 and cw_2 both point in the direction $-a$, and set all remaining coordinates to 0. Then $w \in \mathbb{D}_2^\infty$ and $a + bw_1 + cw_2 = 0$. This proves the criterion. $\square$

Corollary 1. *The function $f_0(z) = z + z^2 + z^3$ has no zeros in $\mathbb{D} \setminus \{0\}$, but $\mathfrak{B}f_0$ vanishes on $\mathbb{D}_2^\infty$.*

Proof. We have $f_0(z) = z(1 + z + z^2)$, and the zeros of $1 + z + z^2$ are the primitive third roots of unity, which lie on $\partial\mathbb{D}$. On the other hand,

$$\mathfrak{B}f_0(w) = 1 + w_1 + w_2,$$

so $\mathfrak{B}f_0(-1/2, -1/2, 0, \dots) = 0$. The point $(-1/2, -1/2, 0, \dots)$ belongs to $\mathbb{D}_2^\infty$. $\square$

Theorem 2 (Zero incomparability). *The two zero conditions*

$$0 \notin f(\mathbb{D} \setminus \{0\}) \quad \text{and} \quad 0 \notin \mathfrak{B}f(\mathbb{D}_2^\infty)$$

are logically independent, even for polynomials in $H_0^2(\mathbb{D})$.

Proof. The preceding corollary gives one direction: $f_0(z) = z + z^2 + z^3$ has no zero in $\mathbb{D} \setminus \{0\}$, but its Bohr lift vanishes at $(-1/2, -1/2, 0, \dots)$.

For the reverse direction, set

$$g(z) = z - z^2 - z^3 + z^6.$$

The prime factorizations $1 = 1$, $2 = p_1$, $3 = p_2$, and $6 = p_1 p_2$ give

$$\mathfrak{B}g(w) = 1 - w_1 - w_2 + w_1 w_2 = (1 - w_1)(1 - w_2).$$

Since $|w_1| < 1$ and $|w_2| < 1$ for $w \in \mathbb{D}_2^\infty$, neither factor can vanish. Hence $\mathfrak{B}g$ is non-vanishing on $\mathbb{D}_2^\infty$.

On the other hand $g(z) = zh(z)$, where

$$h(z) = 1 - z - z^2 + z^5.$$

Now $h(0) = 1$, while

$$h(3/4) = 1 - \frac{3}{4} - \frac{9}{16} + \frac{243}{1024} = -\frac{77}{1024} < 0.$$

By the intermediate value theorem, h has a real zero in $(0, 3/4)$. Thus g has a zero in $\mathbb{D} \setminus \{0\}$, although $\mathfrak{B}g$ is non-vanishing on $\mathbb{D}_2^\infty$. $\square$

Verification Notes

These are finite-dimensional calculations inside the Bohr transform. No computer search was needed. The first proof uses only the prime factorizations $1 = 1$, $2 = p_1$, and $3 = p_2$, and the fact that the first two coordinates of $\mathbb{D}_2^\infty$ can be chosen independently in the unit disk. The reverse example additionally uses $6 = p_1 p_2$.

The result is partial. It does not classify all $f \in H_0^2(\mathbb{D})$. It shows that even for finite positive-index polynomials, Bohr-lift nonvanishing is a several-variable stability condition rather than ordinary nonvanishing of the one-variable function away from the forced zero at the origin. In particular, the two zero conditions are independent.

Novelty and Literature Check

The lightweight run indexes were searched for arXiv:2409.10224 and core phrases including **Bohr transform**, **Hilbert multidisk**, and **non-vanishing**; no duplicate packet or attempt was found. Web/arXiv searches on 2026-06-28 for matching Bohr-transform nonvanishing phrases, including H_0^2 , found no exact duplicate of these finite-polynomial obstructions. The result should therefore be treated as a small new partial observation, subject to human review.

Human Review Recommendation

Check the source's indexing convention for $H_0^2(\mathbb{D})$ and the Bohr transform. In particular, the coefficient of z^1 becomes the constant term of $\mathfrak{B}f$. Once this convention is confirmed, the results are elementary and likely valid obstructions for Problem 2.23.

References

- [1] Jeet Sampat. *Shift-cyclicity in analytic function spaces*. arXiv:2409.10224, 2024.